Commit-derived tasks (GSO)

CodeBase
- llama_cpp/
- examples/

git commit
Faster IQ1 mat-vec using BMI2 instructions

LLM

Generated Perf Test

```
llm = Llama("qwen2-7b.gguf")
prompts = load("sharegpt.json")
def experiment():
  return llm(prompts)
def correct(ref, cur):
  for r, c in zip(ref, cur):
    assert r["toks"] == c["toks"]
```

SWE Agent

Model Patch
src/ggml-cpu.c +80 −15
src/ggml-impl.h +9 −4

Reference Patch
src/ggml-cpu.c +216 −86
cpu-quants.c +97 −12

Test ID	Reference time	Model time
1 ✅	2.17 s	2.94 s
2 ✅	1.9 s	1.62 s
3 ✅	2.74 s	3.10 s

Workload-defined tasks (Ours)

CodeBase su2code/SU2
- SU2_CFD/
- Common/
- case/
- build.sh

Supercomputing competitions
SU2
Airfoil flow simulation
ASC 2014 · SU2
ASC 2016 · ABySS
ASC 2026 · QiboTN
SC 2024 · ICON

Common/src/CConfig.cpp
SU2_CFD/.../CEulerSolver.cpp
CSinglezoneDriver.cpp

```
while (TimeIter < config_container[ZONE_0]
       ->GetnTime_Iter()) {
  Preprocess(TimeIter);
  Run();
  Postprocess();
  Update();
  Monitor(TimeIter);
  Output(TimeIter);
  if (StopCalc) break;
  TimeIter++;
}
```

case/inv_NACA0012.cfg

build.sh

```
set -euo pipefail
cd /app/SU2
if [ ! -d build ]; then
  ./meson.py setup build \
    --prefix=/app/install -Dwith-mpi=disabled \
    -Denable-tecio=false -Denable-cgns=false \
    -Denable-autodiff=false \
    -Denable-directdiff=false \
    -Denable-pywrapper=false -Dcpu-arch=native
fi
./ninja -C build -j4 SU2_CFD/src/SU2_CFD
mkdir -p /app/bin
cp build/SU2_CFD/src/SU2_CFD /app/bin/SU2_CFD
```

mesh_NACA0012_inv.su2

instruction.md

Optimize SU2 source and configuration.

Preserve the full flow solution.

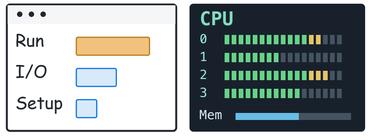
10,216 mesh elements

250 iterations · 5 output snapshots

```
SOLVER= EULER
MACH_NUMBER= @MACH@
AOA= @AOA@
```

SWE Agent

Diagnose



Time breakdown CPU monitor

Implement + integrate

CSysMatrix.hpp
meson.build
CSysMatrix.cpp
inv_NACA0012.cfg
LINEAR_SOLVER_PREC= LU_SGS
+ LINEAR_SOLVER_PREC_THREADS= 1

Validate · Public OJ

Build → Full solve → Check

Public feedback

Retain + select

version.01
version.02
pt1.cfg +2 −0

version.submit

CSysMatrix.cpp	+176	−34
CSysMatrix.hpp	+38	−0
meson.build	+1	−1
inv_NACA0012.cfg	+1	−0

submitted

Separate verifier

SU2 · DeepSeek V4 Pro · 3 attempts

Attempt	Baseline (s)	Optimized (s)	Compile (s)	Agent (h)	Reward
1 ✅	21.35	9.74	172.48	8.00	0.544
2 ✅	21.36	16.18	169.03	4.91	0.242
3 ✅	21.35	8.50	172.15	4.84	0.602